AI ASSISTED CODING

Assignment-19.1

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Task 1 – Python to Java Conversion

Task:

Translate a simple Python class into Java using AI assistance.

Instructions:

- Prompt AI to convert a Python class with attributes and methods into equivalent Java code.
- Verify that constructors, method signatures, and data types are properly handled.
- Run both versions (Python & Java) to compare expected outputs.

Expected Output:

- Equivalent working Java class translated from Python.

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Task 1: Python to Java Conversion

First, here's a sample Python class:

[15] ✓ 0s

```python
class Person:
    def __init__(self, name, age):
        self.name = name
        self.age = age

    def get_name(self):
        return self.name

    def get_age(self):
        return self.age

    def celebrate_birthday(self):
        self.age += 1
        print(f"{self.name} is now {self.age} years old!")

    def describe(self):
        return f"Name: {self.name}, Age: {self.age}"


# Example usage of the Python class
print("--- Python Class Example ---")
person1 = Person("Alice", 30)
print(person1.describe())
person1.celebrate_birthday()
print(person1.describe())

person2 = Person("Bob", 25)
print(person2.describe())
```

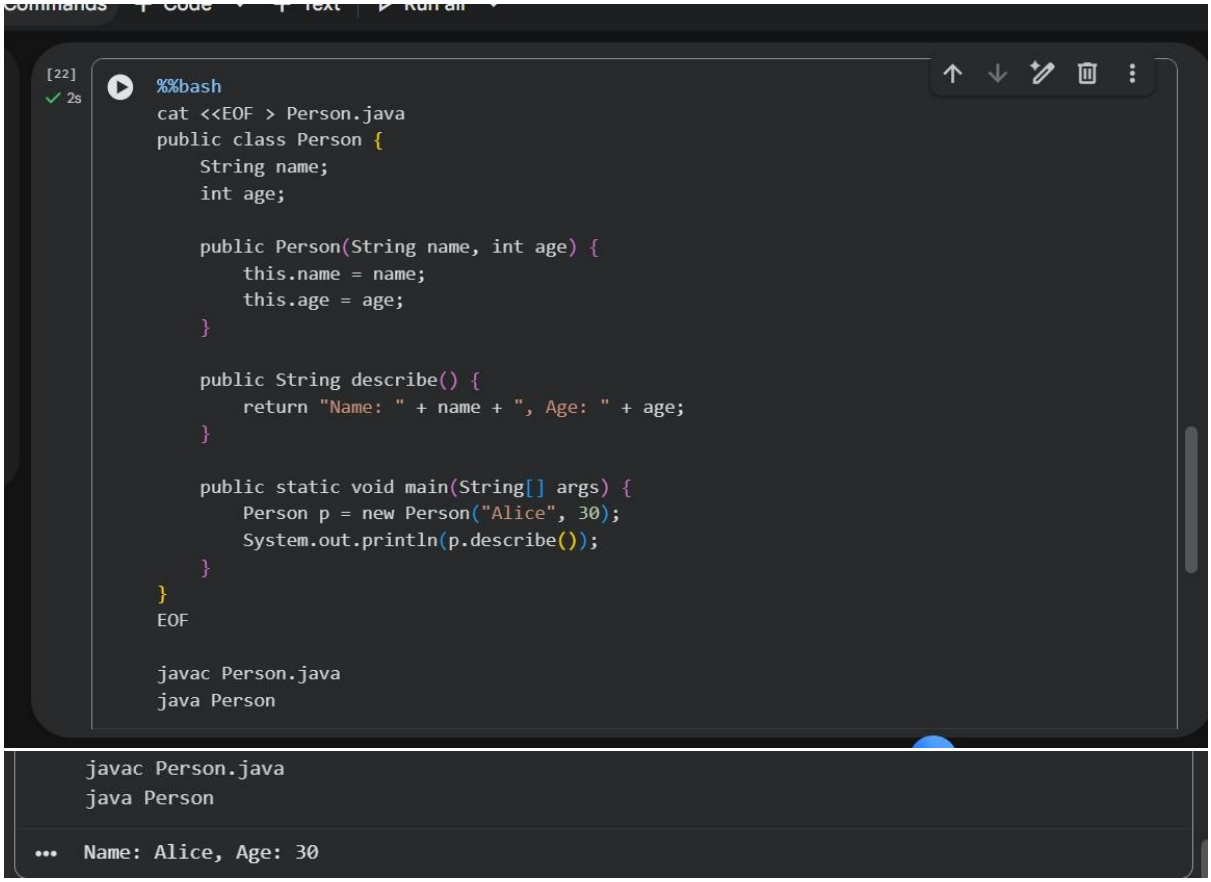
```
--- Python Class Example ---
Name: Alice, Age: 30
Alice is now 31 years old!
Name: Alice, Age: 31
Name: Bob, Age: 25
```

AI-Generated Java Equivalent

Here's the equivalent Java code for the `Person` class. Key differences include explicit type declarations, `public` access modifiers, a constructor with the class name, `this` keyword for instance variables, and standard `getter` and `setter` patterns.

To run this Java code:

1. Save the code below as `Person.java` in a file on your local machine.
2. Open a terminal or command prompt.
3. Navigate to the directory where you saved `Person.java`.
4. Compile the Java code: `javac Person.java`
5. Run the compiled code: `java Person`



Task 2 – C++ to Python Function Conversion

Task:

Convert a C++ function for factorial calculation into Python.

Instructions:

• Ask AI to translate a recursive C++ factorial function into Python.

• Validate correctness by testing with sample inputs.

• Ensure Python version follows Pythonic conventions.

Expected Output:

• A correct Python implementation equivalent to the original

C++ factorial function.

[23]
✓ 1s

```
%%bash
cat <<EOF > factorial.cpp
#include <iostream>

// Recursive function to calculate factorial
long long factorial(int n) {
    if (n == 0 || n == 1) {
        return 1;
    } else {
        return n * factorial(n - 1);
    }
}

int main() {
    int num = 5;
    std::cout << "Factorial of " << num << " is " << factorial(num) << std::endl;

    num = 10;
    std::cout << "Factorial of " << num << " is " << factorial(num) << std::endl;
    return 0;
}
EOF

g++ factorial.cpp -o factorial
./factorial
```

```
Factorial of 5 is 120
Factorial of 10 is 3628800
```

AI-Generated Python Equivalent

Here's the Python equivalent of the recursive factorial function. Key differences include:

- No explicit type declarations for parameters or return values.
- Function definition uses `def` instead of a return type.
- Conditional statements use `if` / `elif` / `else` with colons and indentation.
- The `main` function concept is replaced by direct function calls or a `if __name__ == '__main__':` block.

```
[24]
✓ 0s
def factorial(n):
    """Calculates the factorial of a non-negative integer recursively."""
    if n < 0:
        raise ValueError("Factorial is not defined for negative numbers.")
    elif n == 0 or n == 1:
        return 1
    else:
        return n * factorial(n - 1)


# Validate correctness with sample inputs
print("--- Python Function Example ---")

num1 = 5
result1 = factorial(num1)
print(f"Factorial of {num1} is {result1}")

num2 = 10
result2 = factorial(num2)
print(f"Factorial of {num2} is {result2}")

num3 = 0
result3 = factorial(num3)
print(f"Factorial of {num3} is {result3}")
```

```
[24]
✓ 0s
print(f"Factorial of {num1} is {result1}")

num2 = 10
result2 = factorial(num2)
print(f"Factorial of {num2} is {result2}")

num3 = 0
result3 = factorial(num3)
print(f"Factorial of {num3} is {result3}")

try:
    factorial(-1)
except ValueError as e:
    print(f"Error for negative input: {e}")


--- Python Function Example ---
Factorial of 5 is 120
Factorial of 10 is 3628800
Factorial of 0 is 1
Error for negative input: Factorial is not defined for negative numbers.
```

Task 3 – SQL to Pandas Query

Task:

Translate a SQL query into a Pandas DataFrame operation in

Python.

Instructions:

- Provide AI with a SQL query (e.g., SELECT name, salary FROM employees WHERE salary > 50000).
- Ask AI to generate equivalent Pandas code.
- Test the generated code on a sample DataFrame.

Expected Output:

- Equivalent Pandas query that retrieves the correct subset of data.

```python
# Equivalent Pandas code for:
# SELECT name, salary FROM employees WHERE salary > 50000 ORDER BY salary DESC;

# 1. Filter rows where salary > 50000
filtered_employees = employees_df[employees_df['salary'] > 50000]

# 2. Select 'name' and 'salary' columns
selected_columns = filtered_employees[['name', 'salary']]

# 3. Order by salary in descending order
pandas_result = selected_columns.sort_values(by='salary', ascending=False)

print("\nResulting Pandas DataFrame (equivalent to SQL query):")
display(pandas_result)
```

```
Resulting Pandas DataFrame (equivalent to SQL query):
```

	name	salary
2	Charlie	75000
5	Frank	65000
0	Alice	60000
3	David	52000

Next steps: Generate code with pandas_result | New interactive sheet

Task 4: Multi-Language Snippet Converter - Python to JavaScript: Reverse String

We will translate a simple Python function that reverses a string into its JavaScript equivalent.

[27] ✓ 0s

```python
# Python function to reverse a string
def reverse_string_python(s):
    return s[::-1]

# Test the Python implementation
python_test_string = "Hello, Colab!"
reversed_python_string = reverse_string_python(python_test_string)
print(f"Original Python String: '{python_test_string}'")
print(f"Reversed Python String: '{reversed_python_string}'")
```

```
Original Python String: 'Hello, Colab!'
Reversed Python String: '!balo C ,olleH'
```

[29] ✓ 0s

```javascript
%%javascript
// JavaScript function to reverse a string
function reverseStringJavaScript(s) {
    return s.split('').reverse().join('');
}

// Test the JavaScript implementation
const jsTestString = "Hello, Colab!";
const reversedJsString = reverseStringJavaScript(jsTestString);
console.log(`Original JavaScript String: '${jsTestString}'`);
console.log(`Reversed JavaScript String: '${reversedJsString}'`);
```

...